AP PRECALCULUS

Unit 1 • Lesson 1.12 practice

Lesson 1.12 • Transformations of Functions

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

How is the graph of $g(x) = f(x+3) - 1$ obtained from the graph of f?

(A) Right 3, down 1
(B) Left 3, down 1
(C) Left 3, up 1
(D) Right 3, up 1

Question 2

NO CALCULATOR

How is the graph of $g(x) = f(2x)$ obtained from the graph of f?

(A) Vertical dilation by a factor of 2
(B) Horizontal dilation by a factor of 2
(C) Horizontal dilation by a factor of $\frac{1}{2}$
(D) Vertical dilation by a factor of $\frac{1}{2}$

Question 3

NO CALCULATOR

The point $(4, -2)$ is on the graph of f. Which point must be on the graph of $y = 3f(x-1) + 2$?

(A) $(3, -4)$
(B) $(5, -6)$
(C) $(5, -4)$
(D) $(4, -4)$

Question 4

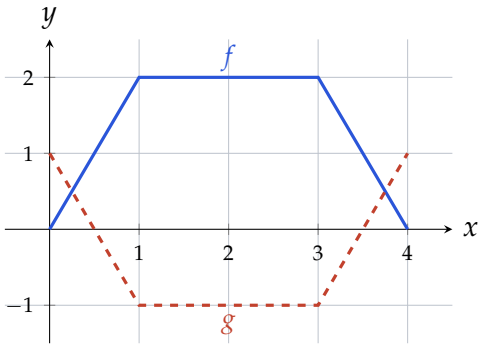
NO CALCULATOR

How is the graph of $y = -f(x)$ obtained from the graph of f?

(A) Reflection over the x-axis
(B) Reflection over the y-axis
(C) Reflection over the line $y = x$
(D) Shift down 1 unit

Question 5 (graph) NO CALCULATOR

The graphs of f (solid) and g (dashed) are shown. Which is an equation for g in terms of f?



(A) $g(x) = -f(x) - 1$
(B) $g(x) = f(-x) + 1$
(C) $g(x) = -f(x) + 1$
(D) $g(x) = -f(x + 1)$

Question 6 (Free response) NO CALCULATOR

Let $f(x) = x^2$ and $g(x) = -2(x-1)^2 + 8$.

(a) Describe the transformations that take the graph of f to the graph of g.
(b) Find the zeros of g.
(c) If f is restricted to the domain $[-2, 2]$ (range $[0, 4]$), find the corresponding domain and range of g.